Technical specification template: Earthing, Bonding and Lightning Protection Interface

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A5. Generated 2026-11-05 07:50 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for earthing, bonding and lightning protection interface for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEEE 80 (substation grounding), as a method reference
- IEC 60364-5-54 (earthing arrangements and protective conductors)
- IEC 62305 series (lightning protection)
- Local utility and authority requirements [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit study pending utility, transformer	NOT CALCULATED

	and generator data]	

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Earthing system type	TN-S	UNVERIFIED_PLACEHOLDER. Assumed for the sample
Soil resistivity	[From a site soil resistivity survey, Wenner method]	Pending site data
Grid geometry	[From the earthing design]	Pending
Earth-fault current and clearing time	[From the fault and protection studies]	Pending
Touch and step voltage limits	[Per the adopted method]	Pending

5. Technical requirements

5.1 The earthing design shall be based on a soil resistivity survey and shall demonstrate that touch and step voltages and the ground potential rise are within the limits of the adopted method.

5.2 The design shall state the earth-fault current, its split between the grid and other paths, and the clearing time used.

5.3 Bonding conductors, connections, test points and corrosion protection: [project to define].

5.4 Interfaces with the lightning protection system, structural steel, the utility earth and IT equipment earthing: [project to define].

6. Manufacturer data required

The tenderer shall supply the datasheets and values below with its offer, each with the document name and revision. A value without a source document is not accepted. These are also tracked in the manufacturer-data register and are imported into the calculations through the data override mechanism.

Parameter id	Parameter	Datasheet	Current status
earthing_study.grid_area_m2	Earth grid area	Earthing grid layout drawing and design	MISSING
earthing_study.grid_conductor_length_m	Total buried grid conductor length	Earthing grid layout drawing and design	MISSING
earthing_study.grid_burial_depth_m	Grid burial depth	Earthing grid layout drawing and design	MISSING
earthing.earthing_system_type	Earthing system type	Project earthing design basis	UNVERIFIED_PLACEHOLDER
earthing.soil_resistivity	Soil resistivity	Site soil resistivity survey (Wenner method) report	MISSING

7. Inspection and quality requirements

7.1 Review of the survey report, the grounding study and the drawings.

7.2 Inspection of buried conductors and connections before backfill, with records.

7.3 Routine tests on every unit as required by the applicable standard and this specification.

7.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

7.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

7.6 Calibration certificates for test instruments.

8. Factory acceptance test (FAT) requirements

8.1 Not applicable to the installation. Factory tests apply only to manufactured components (for example exothermic moulds and test links): [project to define].

8.2 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9. Commissioning requirements

9.1 Continuity of all bonds; earth resistance measurement (fall-of-potential or the specified method).

9.2 Verification of touch and step voltage against the design where specified; records retained.

9.3 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

10. Submittals and documentation

- Datasheets, drawings and type-test reports.
- Inspection and test plan, FAT procedure and report.
- Installation, operation and maintenance manuals.
- Spare parts and special tools lists.
- Warranty terms [project to define].

11. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- This template has not been reviewed by an independent engineer.